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Cost & Performance Trade-offs

Token Economics • GPU Hours • Batching Strategies

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Masai School | IITP-AIML-2506 | Week 34 — Session 2/2

04 April 2026



The $72,000 Wake-Up Call

The Problem

A startup built a customer support bot using GPT-4 with 8K context.
Prototype cost seemed harmless. Production cost was devastating.

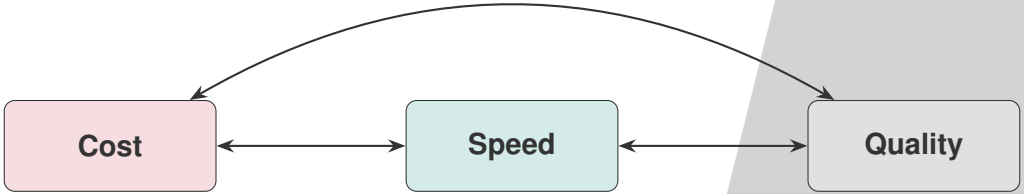
- **Prototype:** 100 calls/day = $12/day — seemed cheap
- **After launch:** 200,000 calls/day = $24,000/day
- **Monthly bill:** $72,000 — more than their entire cloud budget
- **Fix:** Switched to GPT-3.5 Turbo + smart prompting → **90% cost reduction**

Takeaway

This is not hypothetical. Cost optimization is a **critical engineering skill**.

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Why This Lecture Matters



- **Money:** GPT-4 can cost 100x more than GPT-3.5 per call
- **Latency:** Users abandon chatbots that take $>$3 seconds
- **Sustainability:** GPU compute has a carbon footprint
- **Career:** Every ML engineer needs to answer “how much will this cost?”

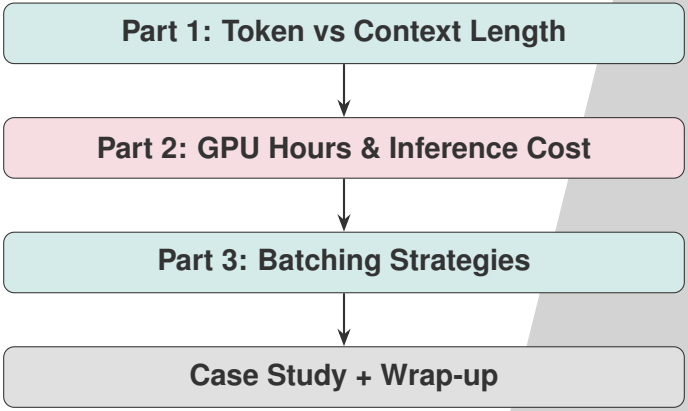
What You Will Learn Today

Learning Objectives

By the end of this session, you will be able to:

1. Understand how **tokens and context length** drive LLM costs
2. Calculate **GPU requirements** and inference costs for different model sizes
3. Explain **batching strategies** and their impact on throughput
4. Apply all three levers to **optimize** a real-world LLM deployment
5. **Estimate monthly costs** for production LLM systems

Today’s Roadmap



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The LLM Pricing Landscape

Key Insight

You always pay for compute — the question is **how directly**.

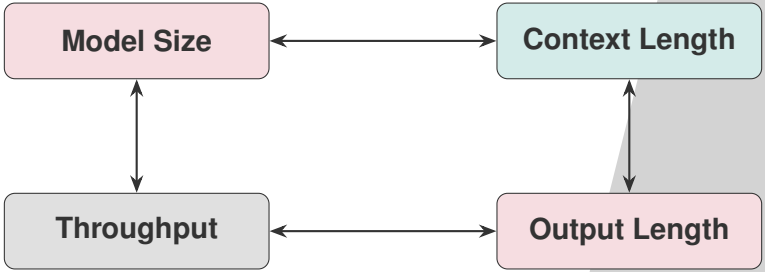
Model	How You Pay	Example
API Providers	Per token (input + output)	GPT-4 Turbo: $10/M in, $30/M out
Cloud GPU Rental	Per GPU-hour	A100: $3/hr on AWS
Self-Hosted	Hardware + electricity	8x H100 server: $200K upfront

Examples

GPT-3.5 Turbo: $0.50/M input, $1.50/M output — **20x cheaper** than GPT-4 Turbo.

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The Four Cost Drivers



- **Model size:** More parameters = more FLOPS per token
- **Context length:** Attention is $O(n^2)$ — doubling context $\approx$ quadruples compute
- **Throughput demand:** More req/sec = more GPUs needed
- **Output length:** Generation is sequential and typically the bottleneck

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How We Got Here — A Brief History

Year	Milestone
2018	BERT (110M params) — started the compute conversation
2020	GPT-3 (175B) — inference cost becomes a business concern
2020	Scaling Laws (Kaplan et al.) — size vs performance quantified
2022	Chinchilla — “most models are too big and undertrained”
2023	Llama 2 + quantization boom — “run it yourself” becomes viable
2023–24	vLLM, continuous batching — 2–5x serving efficiency gains
2024–25	MoE, speculative decoding — new Pareto frontier

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Tokens — The Atomic Unit of LLM Cost

Key Idea

A token is a sub-word unit, typically 3–4 characters in English.

- “Hello, world!” = **4 tokens**
- Different tokenizers produce **different token counts** for the same text
- Rule of thumb: **1 token** $\approx$ **0.75 English words** (or 1 word $\approx$ 1.33 tokens)
- Non-English text and code often use **more tokens** per semantic unit

Examples

“The quick brown fox” = 4 tokens (GPT-4).
“Pneumonoultramicroscopicsilicovolcanoconiosis” = 10 tokens.

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How Token Pricing Works

Formula

Total cost = (input_tokens $\times$ input_price) + (output_tokens $\times$ output_price)

- Providers charge per **1K or 1M** tokens
- Input and output tokens are priced **separately**
- Output tokens typically cost **2–6x more** than input tokens
- **Hidden cost:** system prompts count as input tokens on *every* call

Examples

GPT-4 Turbo call:
2,000 input + 500 output = (2000 $\times$ \$0.01/1K) + (500 $\times$ \$0.03/1K) = \$0.02 + \$0.015 = **\$0.035**

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Context Length and the Quadratic Cost Curse

The Problem

Self-attention computes pairwise interactions: $O(n^2)$ in sequence length.
Doubling context doesn't double cost — it roughly **quadruples** attention compute.

- Context length = max tokens the model can "see" at once
- Memory usage scales with context: **KV cache grows linearly** per layer
- Longer context = fewer concurrent requests per GPU = **lower throughput**

Context	Attention Ops	Relative Cost
4K tokens	16M	1x
8K tokens	64M	4x
32K tokens	1,024M	64x

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System Prompts — The Silent Budget Killer

Important Caveat

System prompts are prepended to **every** API call.

- A 500-token system prompt on 100K calls/day = **50M extra input tokens/day**
- At GPT-4 Turbo pricing: 50M $\times$ \$0.01/1K = **\$500/day** just for system prompts
- **Prompt caching** (offered by some providers) can reduce this dramatically
- Best practice: keep system prompts **concise**, use caching when available

Examples

Chatbot with 800-token system prompt at 100K calls/day:
800 $\times$ 100K $\times$ 365 $\times$ \$10/1M = **\$292K/year** just on the system prompt.

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Prompt Engineering — Not Just for Quality

Key Insight

Shorter prompts = fewer input tokens = lower cost. Same principle for outputs.

- Use **structured formats** (JSON, bullets) instead of verbose prose
- “Answer in 50 words or less” — constraining output saves on output tokens
- Few-shot examples are expensive — use **1–2 instead of 5–6** when possible
- Chain-of-thought is powerful but costly — use **selectively**

Verbose (28 tokens)

“Please analyze the following text and provide a comprehensive summary covering all main points...”

Optimized (5 tokens)

“Summarize in 3 bullets:”

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When You Can’t Fit It All — Truncation & Chunking

- Not all input is equally important — **truncate strategically**
- **Recency bias:** keep recent turns, summarize older ones
- **Chunking:** split long documents, process chunks independently, merge results
- **Map-reduce:** summarize each chunk, then summarize the summaries
- **Sliding window:** overlapping chunks to avoid losing context at boundaries

Examples

50-page legal document:
Full context: 60K tokens $\rightarrow$ \$0.60/call.
Chunk + map-reduce: 5 chunks $\times$ 4K + 1 merge of 2K = 22K tokens $\rightarrow$ \$0.22/call.
63% cheaper.

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Token Cost Across Popular Models (2024–25 Pricing)

Model	Input/1M	Output/1M
GPT-4 Turbo	$10.00	$30.00
GPT-4o	$2.50	$10.00
GPT-3.5 Turbo	$0.50	$1.50
Claude 3.5 Sonnet	$3.00	$15.00
Llama 3 70B (self-hosted)	~$0.50	~$0.50
Llama 3 8B (self-hosted)	~$0.05	~$0.05

The Spread

1M tokens/day workload: GPT-4 Turbo = **$40/day** vs Self-hosted Llama 3 8B = **$0.10/day**.
That is a **400x** difference.

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Quiz 1, Q1: Token Math

Question

Your chatbot uses a 600-token system prompt. You handle 50,000 conversations/day, each with 4 API calls. Using GPT-4 Turbo ($10/1M input tokens), how much do you spend **per month** just on system prompt tokens?

- A) $36
- B) $360
- C) $3,600
- D) $36,000

Take 1 minute to calculate, then we'll reveal the answer.

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Quiz 1, Q1: Answer

Answer

D) $36,000

Step-by-Step

1. Tokens per day: 600 $\times$ 50,000 $\times$ 4 = **120M tokens/day**
2. Monthly: 120M $\times$ 30 = **3,600M = 3.6B tokens**
3. Cost: 3,600 $\times$ $10/1M = **$36,000/month**

Takeaway: A single system prompt, at scale, can cost more than your rent.

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Quiz 1, Q2: Context Length Impact

Question

If self-attention cost scales as $O(n^2)$, and processing 4K tokens costs X compute units, approximately how many compute units does processing 16K tokens require?

- A) 2X
- B) 4X
- C) 8X
- D) 16X

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Quiz 1, Q2: Context Length Impact

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If self-attention cost scales as $O(n^2)$, and processing 4K tokens costs X compute units, approximately how many compute units does processing 16K tokens require?

- A) 2X
- B) 4X
- C) 8X
- D) 16X

Answer

D) 16X

16K / 4K = 4x more tokens. Since $O(n^2)$: $4^2 = 16X$ compute.

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Quiz 1, Q3: Cost Optimization

Question

Which of these strategies would **NOT** reduce your token costs?

- A) Replacing 5-shot prompts with 1-shot prompts
- B) Adding “Answer in 100 words or less” to your prompt
- C) Switching from GPT-4 to GPT-3.5 Turbo
- D) Increasing the temperature parameter from 0.7 to 1.0

Quiz 1, Q3: Cost Optimization

Question

Which of these strategies would **NOT** reduce your token costs?

- A) Replacing 5-shot prompts with 1-shot prompts
- B) Adding “Answer in 100 words or less” to your prompt
- C) Switching from GPT-4 to GPT-3.5 Turbo
- D) Increasing the temperature parameter from 0.7 to 1.0

Answer

D) Temperature affects randomness, not token count or pricing.

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5-Minute Break

Coming Up Next

Part 2: GPU Hours & Inference Cost

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Know Your GPUs — A Spotter's Guide

GPU	VRAM	Cost/hr	Best For
T4	16 GB	$0.35	Small models ($\leq$7B INT8)
A10G	24 GB	$1.00	7B–13B models
A100	80 GB	$3.00	13B–70B models
H100	80 GB	$4.50	High throughput

Key Metric

Not raw FLOPS, but **tokens-per-second-per-dollar**.

Examples

Llama-2 7B INT8: T4 = 15 tok/s ($0.006/1K), A10G = 35 tok/s ($0.008/1K).
T4 is cheapest per token for this model.

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The Size-Cost-Quality Triangle

- Bigger models score higher but **cost more to run**
- The relationship is **NOT linear**:
  - 70B is $\sim$10x more expensive than 7B
  - But often only 10–20% better on benchmarks
- **Key question:** Does your use case need 70B quality?
- Classification, extraction, simple QA: **7B often sufficient**
- Complex reasoning, code gen: **70B may be worth it**

Model	MMLU	Cost
Llama 7B	46%	1x
Llama 13B	55%	2x
Llama 70B	69%	10x

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Quantization — Same Model, Fewer Bits, Lower Cost

Key Idea

Reduce the precision of model weights to shrink size and increase speed.

Precision	Size Reduction	Speed Gain	Quality Loss
FP16 (default)	1x	1x	0%
INT8	2x smaller	$\sim$1.5x faster	$<$1%
INT4 (GPTQ/AWQ)	4x smaller	$\sim$2x faster	1–3%

Examples

Llama-2 70B:
FP16 = 140 GB (needs 2x A100) vs INT4 = 35 GB (fits on 1x A100).
Half the GPU cost with minimal quality loss.

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Latency vs Throughput — Two Different Games

Latency

Time for **one** request.
Matters for interactive chat.

Optimize: small batch, fast GPU, small model.
Target: <500ms TTFT.

Throughput

Requests **per second**.
Matters for batch processing.

Optimize: large batch, maximize GPU utilization.
Target: 1000+ docs/hour.

Trade-off

Optimizing for one often **hurts the other**. Choose based on your use case.

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The Inference Cost Equation

Formula

$$\text{Cost per token} = \frac{\text{GPU cost per hour}}{\text{tokens per hour}}$$

$$\text{Monthly cost} = \text{cost per token} \times \text{tokens per month}$$

- tokens_per_hour = tokens_per_second $\times$ 3600
- Depends on: model size, GPU type, batch size, precision, sequence length
- Include: GPU + networking + storage + engineering overhead

Examples

A100 at $3/hr, Llama-2 13B at 200 tok/s:
$3/hr ÷ 720K tok/hr = $0.0042/1K tokens.
At 100M tokens/month: **$420/month**.

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Quiz 2, Q1: GPU Economics

Question

You need to serve a 70B parameter model at FP16 (140 GB VRAM). What is the minimum setup?

- A) 1x T4 (16 GB)
- B) 1x A100 (80 GB)
- C) 2x A100 (160 GB)
- D) 1x H100 (80 GB)

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Quiz 2, Q1: GPU Economics

Question

You need to serve a 70B parameter model at FP16 (140 GB VRAM). What is the minimum setup?

- A) 1x T4 (16 GB)
- B) 1x A100 (80 GB)
- C) 2x A100 (160 GB)
- D) 1x H100 (80 GB)

Answer

C) 140 GB model needs at least 140 GB VRAM.
2x A100 (160 GB) is the minimum option that fits.

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Quiz 2, Q2: Quantization Trade-off

Question

You quantize a 70B model from FP16 to INT4 (GPTQ). What is the approximate impact?

- A) 4x smaller, 2x faster, 0% quality loss
- B) 4x smaller, 2x faster, 1–3% quality loss
- C) 2x smaller, 4x faster, 5–10% quality loss
- D) 2x smaller, 1.5x faster, $<$1% quality loss

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- C) 2x smaller, 4x faster, 5–10% quality loss
- D) 2x smaller, 1.5x faster, $<$1% quality loss

Answer

B) INT4 = 4x compression, $\sim$2x speedup, 1–3% quality drop on benchmarks.

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Quiz 2, Q3: Cost Calculation

Question

An A100 costs $3/hr and serves Llama-2 13B at 200 tokens/sec. What is the cost per 1M tokens?

- A) $0.42
- B) $4.17
- C) $15.00
- D) $42.00

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Quiz 2, Q3: Cost Calculation

Question

An A100 costs $3/hr and serves Llama-2 13B at 200 tokens/sec. What is the cost per 1M tokens?

- A) $0.42
- B) $4.17
- C) $15.00
- D) $42.00

Answer

B) $4.17

200 tok/s $\times$ 3600 = 720,000 tok/hr.

$3 / 720,000 = $0.00000417/token. Per 1M: $4.17.

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Batching — The Most Underrated Cost Lever

Key Idea

Batching = processing multiple requests simultaneously on one GPU.

- Without batching: GPU processes one request, **sits idle** during memory transfers
- With batching: GPU **fills its compute capacity** with multiple requests
- Why it works: LLM inference is often **memory-bandwidth-bound**, not compute-bound
- Adding more requests doesn't proportionally increase per-request latency

Analogy

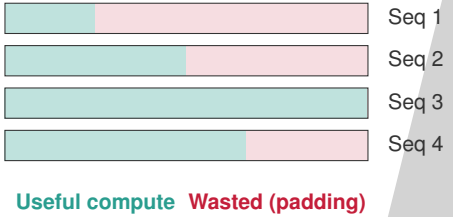
Restaurant analogy: One pizza in a large oven vs filling the oven.
Same time, **8x throughput**.

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Static Batching — The Simple Approach

Problems with Static Batching

1. Must **wait** to collect a full batch (adds latency)
2. All sequences **padded to the longest** one (wastes compute)
3. Entire batch completes together — short sequences **wait** for long ones



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Continuous Batching — The Modern Approach

Method

New requests can **join mid-batch** as old ones finish. No padding. No waiting.

- **Dynamic batching:** varying batch sizes based on incoming request rate
- **Continuous batching** (iteration-level): new requests enter as slots free up
- No padding waste: each sequence is its own length
- Pioneered by **Orca paper**, implemented in **vLLM** and **TGI**

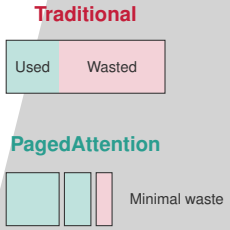
Examples

vLLM with continuous batching: **2–5x throughput improvement** over static batching with the same hardware.

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PagedAttention — Why vLLM Changed Everything

- **Traditional KV cache:** pre-allocate max_seq_len per request — massive waste
- **PagedAttention:** allocates KV cache in **pages** (blocks), like virtual memory
- Reduces memory waste from ∼60% to ∼4%
- Enables **2–4x more concurrent requests** per GPU
- vLLM (UC Berkeley) is the most popular implementation



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The Numbers — What Batching Actually Delivers

Strategy	Req/sec	GPUs for 1K req/s	Cost/hr
No batching	5–10	∼100 A100s	$300
Static (batch=8)	25–40	∼25 A100s	$75
Continuous (vLLM)	50–100	∼10 A100s	$30

Impact

10x throughput = 10x fewer GPUs = **10x cost reduction**.
Savings: $270/hr = **$194K/month**.

Real-world caveat: gains depend on request pattern, sequence lengths, and model size.

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Quiz 3, Q1: Batching Basics

Question

In static batching with batch_size=8, you have 8 sequences of lengths [100, 150, 200, 800, 100, 120, 180, 300]. What % of padded tokens are wasted?

- A) 25%
- B) 50%
- C) 69%
- D) 80%

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Quiz 3, Q1: Batching Basics

Question

In static batching with batch_size=8, you have 8 sequences of lengths [100, 150, 200, 800, 100, 120, 180, 300]. What % of padded tokens are wasted?

- A) 25%
- B) 50%
- C) 69%
- D) 80%

Answer

C) 69%.
All padded to 800. Total = 8 $\times$ 800 = 6,400. Actual = 1,950. Wasted = 4,450/6,400 = 69.5%.

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Quiz 3, Q2: Continuous Batching

Question

What is the key advantage of continuous batching over static batching?

- A) It uses less GPU memory
- B) It produces higher quality outputs
- C) New requests can join the batch as completed ones leave
- D) It eliminates the need for GPUs

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Quiz 3, Q2: Continuous Batching

Question

What is the key advantage of continuous batching over static batching?

- A) It uses less GPU memory
- B) It produces higher quality outputs
- C) New requests can join the batch as completed ones leave
- D) It eliminates the need for GPUs

Answer

C) Continuous batching treats each iteration independently — finished sequences free slots immediately for new requests.

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Quiz 3, Q3: Real-World Batching

Question

A company moves from no batching (5 req/s per GPU) to vLLM continuous batching (50 req/s per GPU). They handle 500 req/s. How many fewer A100s ($3/hr) and monthly savings?

- A) 90 fewer GPUs, $194,400/month
- B) 45 fewer GPUs, $97,200/month
- C) 10 fewer GPUs, $21,600/month
- D) 9 fewer GPUs, $19,440/month

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Question

A company moves from no batching (5 req/s per GPU) to vLLM continuous batching (50 req/s per GPU). They handle 500 req/s. How many fewer A100s ($3/hr) and monthly savings?

- A) 90 fewer GPUs, $194,400/month
- B) 45 fewer GPUs, $97,200/month
- C) 10 fewer GPUs, $21,600/month
- D) 9 fewer GPUs, $19,440/month

Answer

A) Before: 500/5 = 100 GPUs. After: 500/50 = 10. Diff: 90.
Savings: 90 × $3/hr × 720 hrs/month = **$194,400/month**.

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Case Study: From $15K to $1.5K/Month

Scenario

E-commerce chatbot, 50K conversations/day, 6 turns average.

1. **Lever 1 — Tokens:** Compressed system prompt (2K → 400 tokens), added “be concise”.
   Savings: 60%
2. **Lever 2 — Model:** GPT-3.5 Turbo for 80% of queries (simple FAQ), GPT-4 only for complex.
   Savings: 75% on model cost
3. **Lever 3 — Self-hosting:** Moved FAQ tier to Llama-2 13B with vLLM.
   Savings: additional 50%

Result

$15,000/month → $1,500/month with **no measurable quality drop**.

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Final Quiz — Test Your Knowledge

1. If GPT-4 Turbo costs $10/1M input tokens and you send 500M input tokens/month, what is your monthly input cost?
2. A model requires 40 GB VRAM. Can it run on a single T4 (16 GB)?
3. Continuous batching improves throughput by approximately ___x over no batching?
4. INT4 quantization reduces model size by approximately ___x vs FP16?
5. Name two strategies to reduce token costs without changing the model.

Take 5 minutes. Work individually or in pairs.

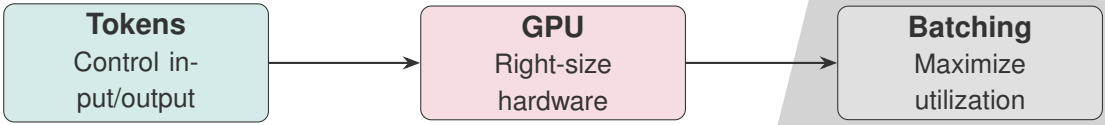
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Final Quiz — Answers

1. 500M $\times$ \$10/1M = **\$5,000/month**
2. **No** — need at least an A10G (24 GB) or 2x T4
3. **5–10x** improvement
4. **4x** size reduction
5. Any two of: compress system prompt, constrain output length, reduce few-shot examples, use chunking/map-reduce, enable prompt caching

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Summary — Three Levers for Cost Optimization



- **95% of quality at 20% of the cost** is almost always achievable
- Tools: vLLM, TGI, OpenAI tokenizer, LiteLLM
- Papers: Scaling Laws (2020), Chinchilla (2022), PagedAttention (2023)
- **Key takeaway:** Cost optimization is not a one-time task — it is an ongoing engineering discipline

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Thank You!

Questions?



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All lecture materials available at:

priyanshsaxena.com/masai-iitp-aiml-2506

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